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The Chalk-Trick

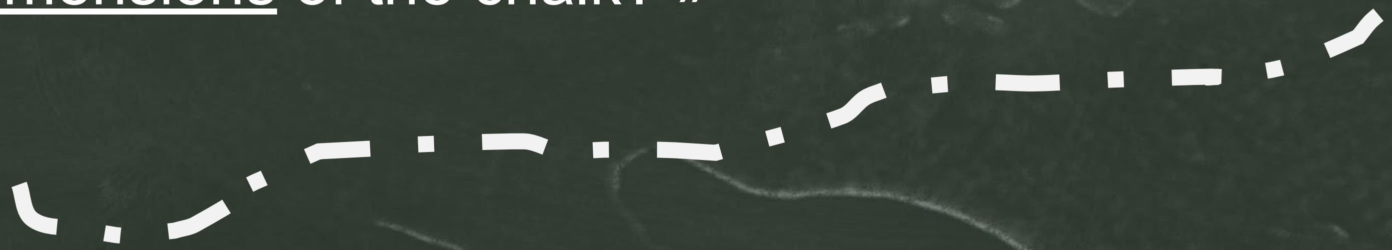
Université Paris Cité

International Physicist Tournament
Problem 11

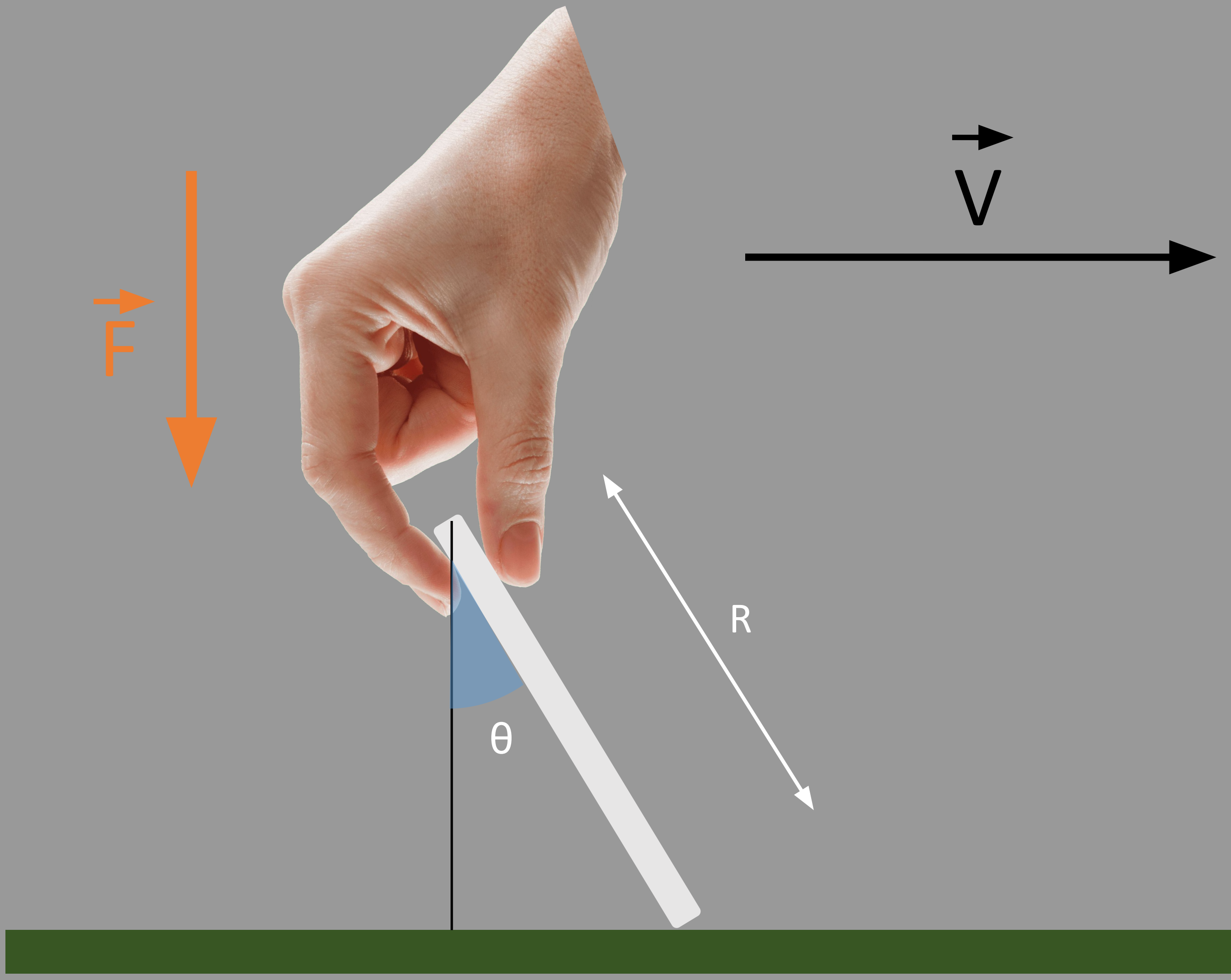
« It is possible to draw continuous lines in a blackboard with chalk.

However, by changing the angle of contact, the line drawn on the board becomes a dotted line, though the movement is still continuous.

What parameters from the relative movement between the chalk and the board can be inferred from the resulting trace? Is it possible to infer anything about the dimensions of the chalk? »



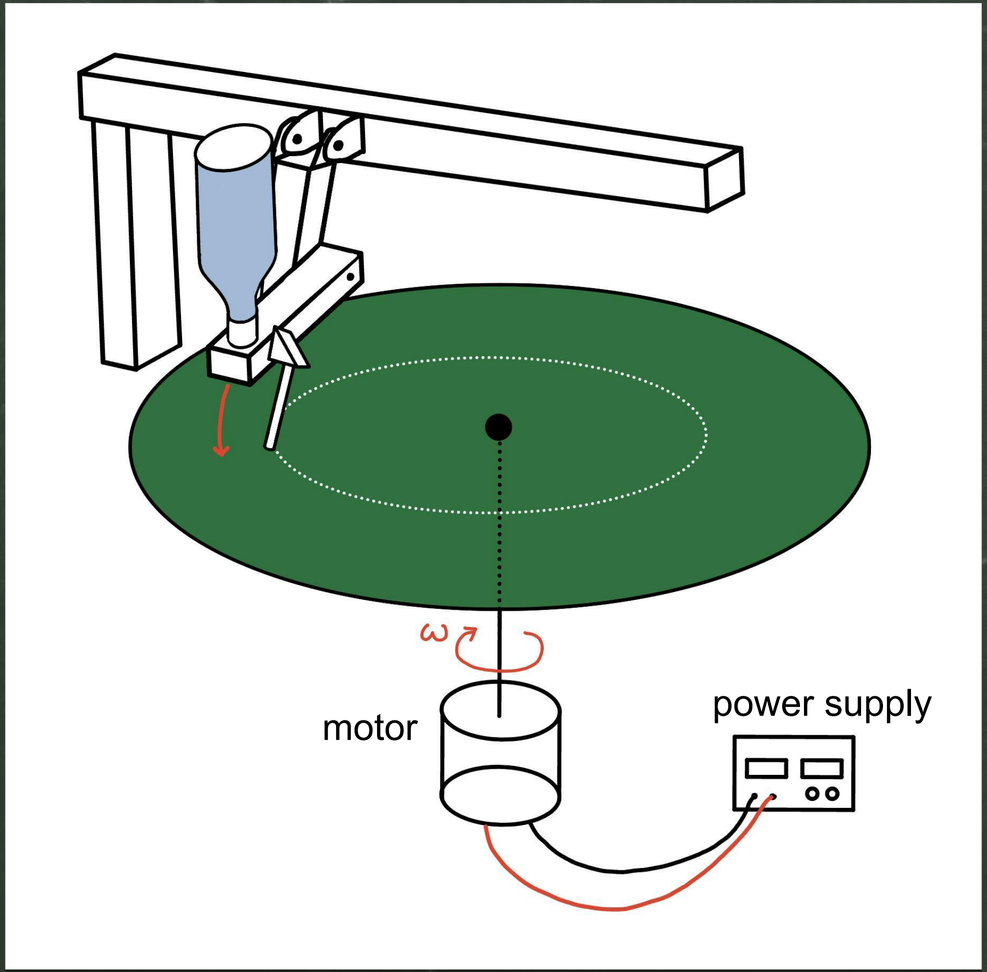

$\vec{A} + \vec{B}$

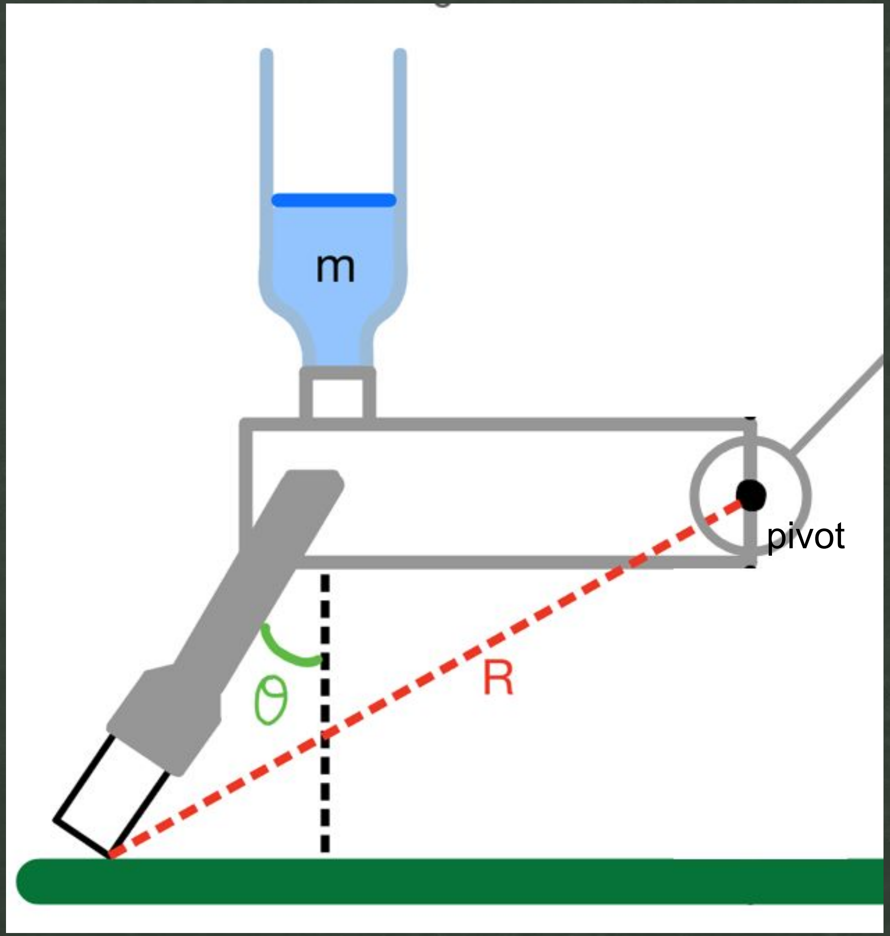


4 parameters :

- Length of the chalk (cm)
- Velocity V (m.s-1)
- Force F (N)
- Angle θ (degrees)

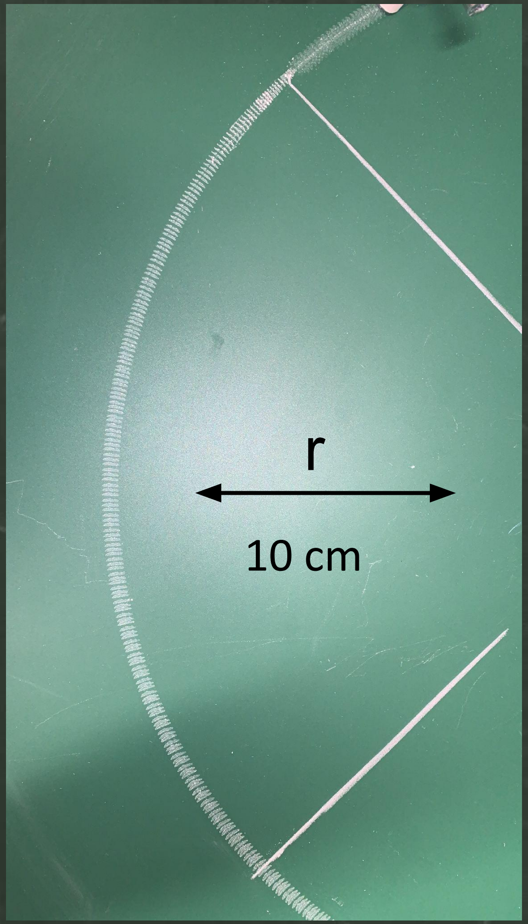
II. Experiment



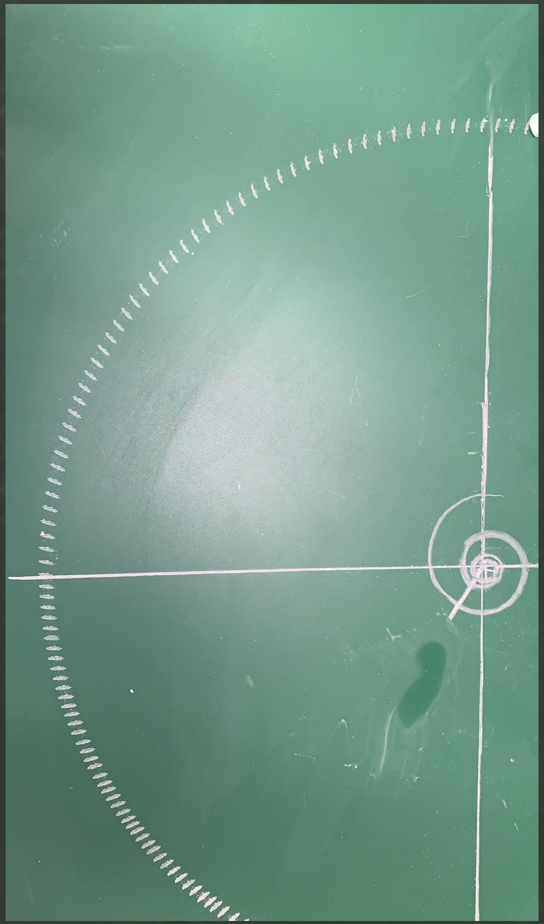


III. Results

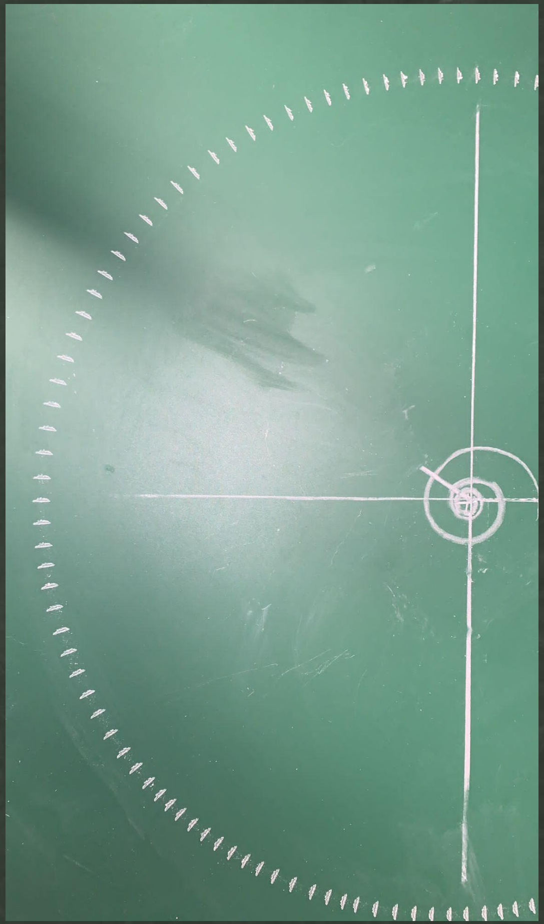
III.A – Movement velocity



0.92 rad/s

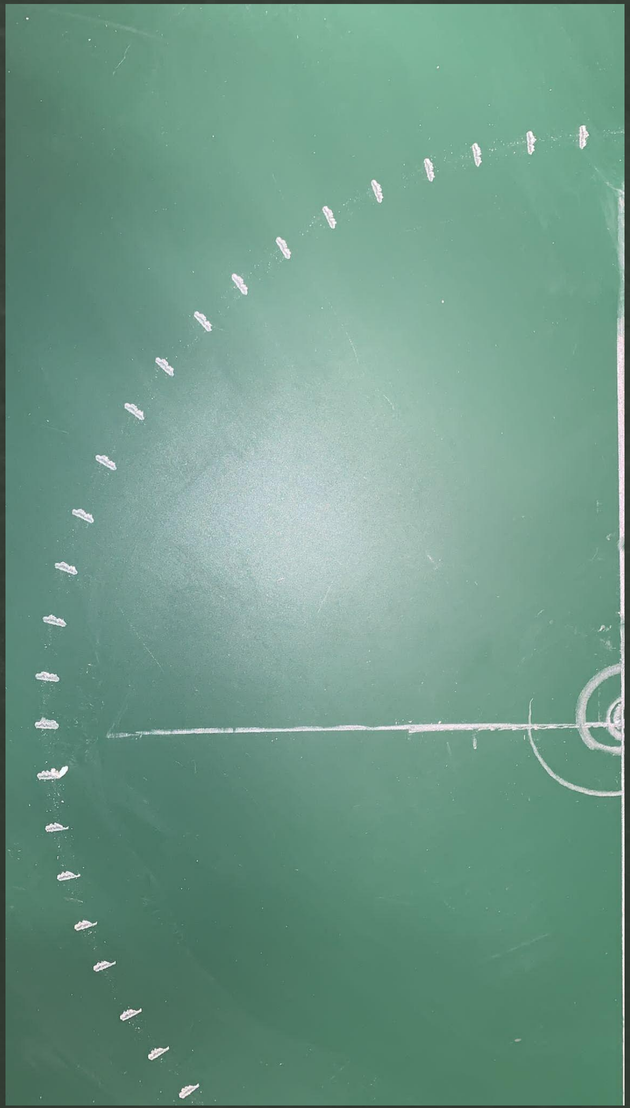


1.84 rad/s

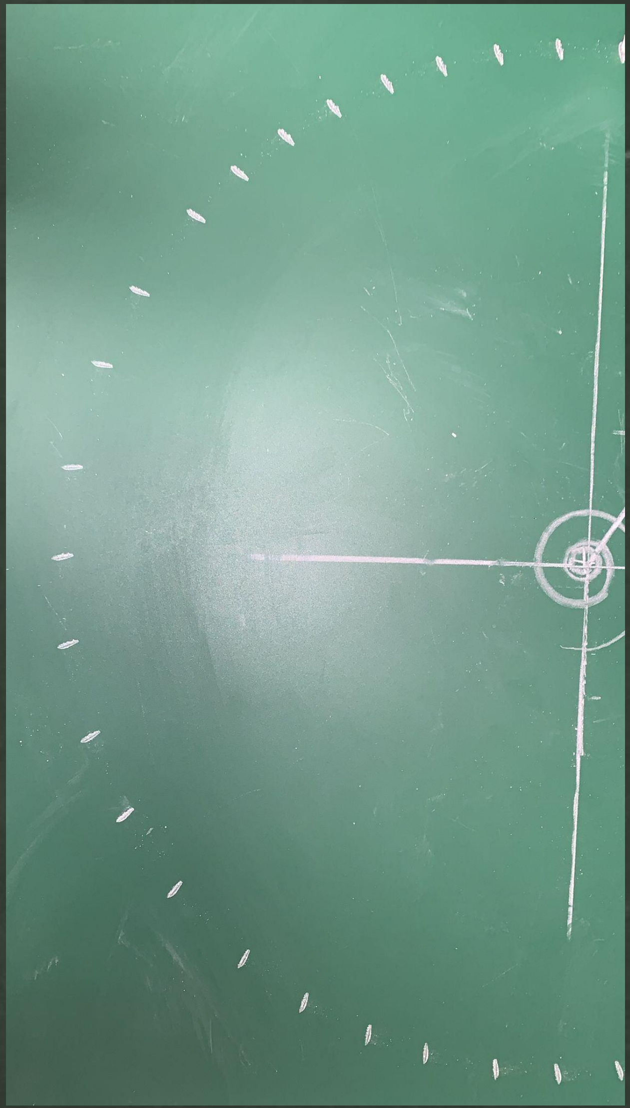


2.76 rad/s

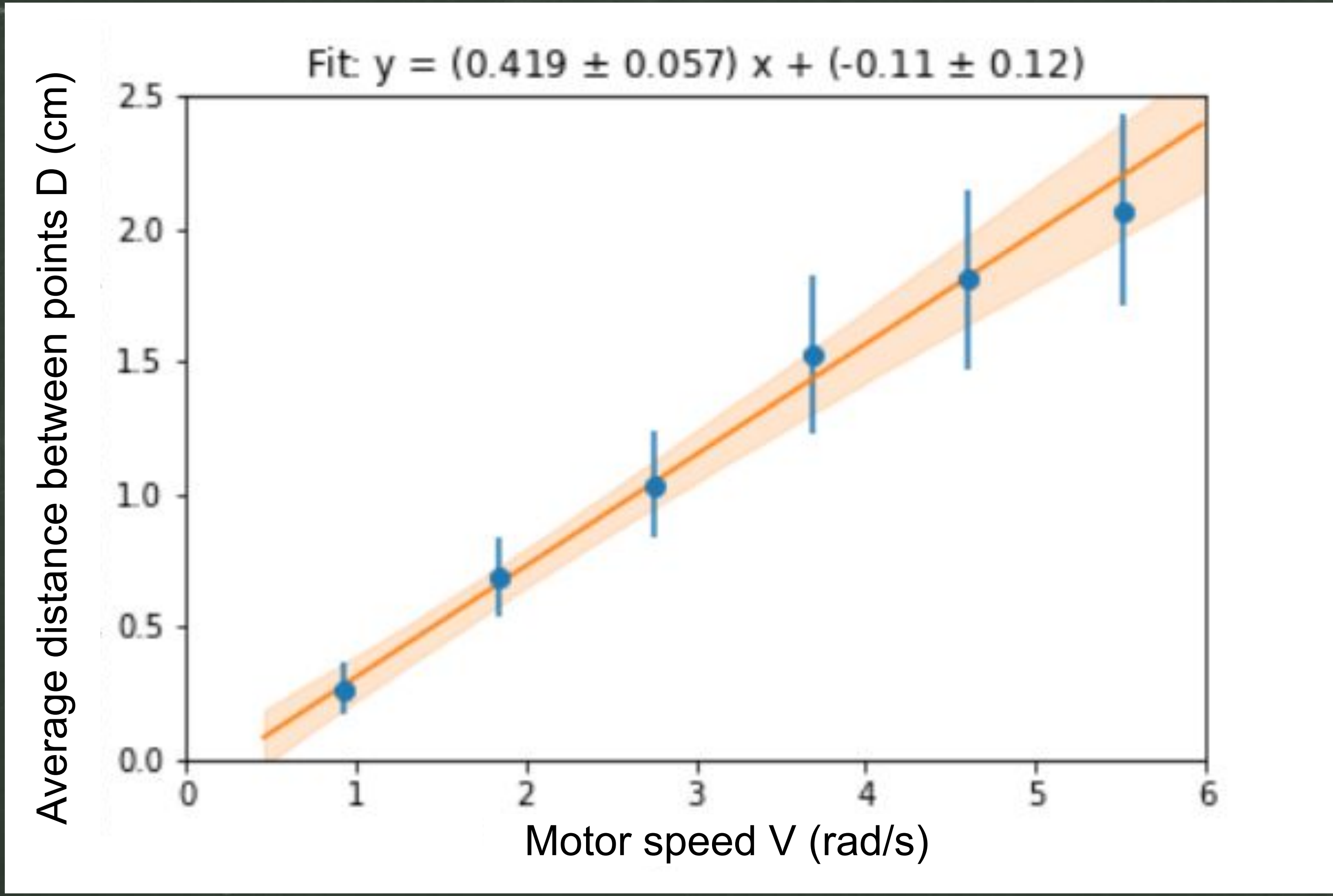
3.68 rad/s



4.60 rad/s



5.52 rad/s

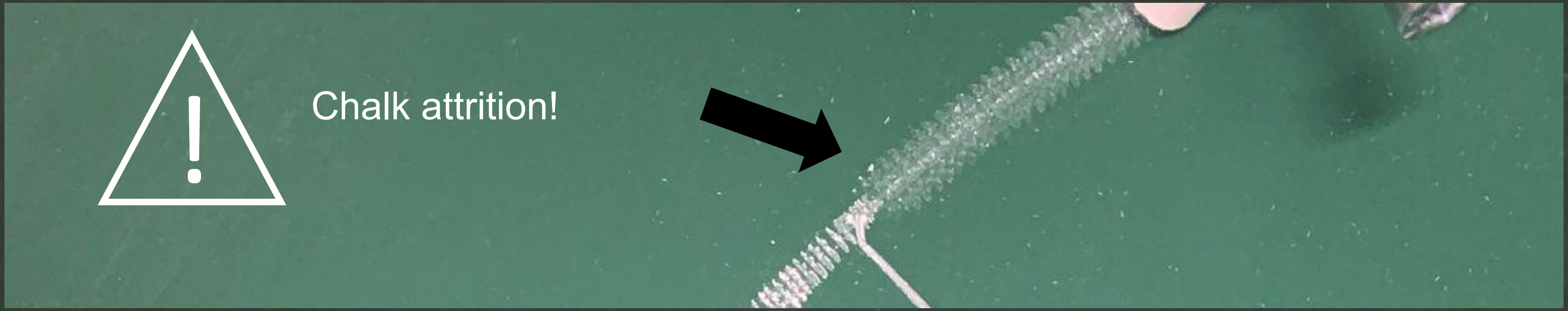


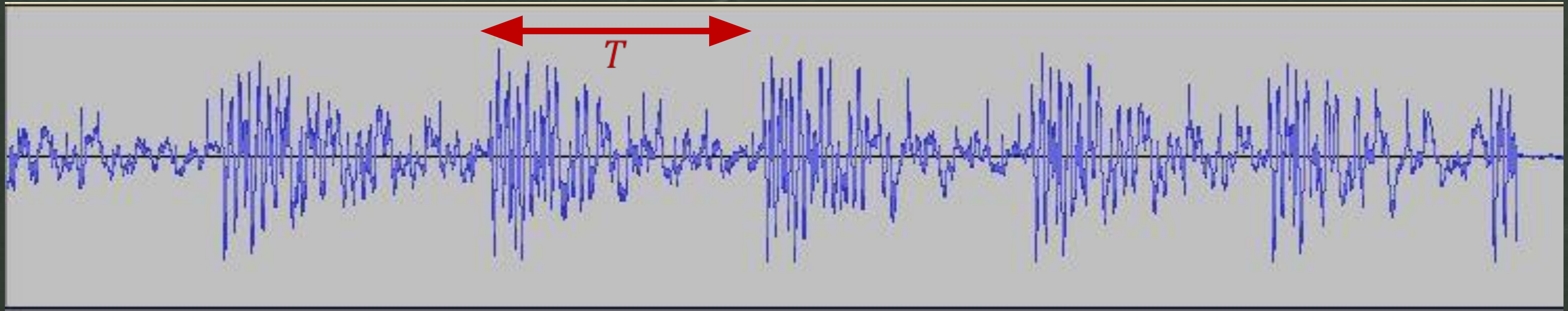
$D = V \times T$

$V = r \times \omega$

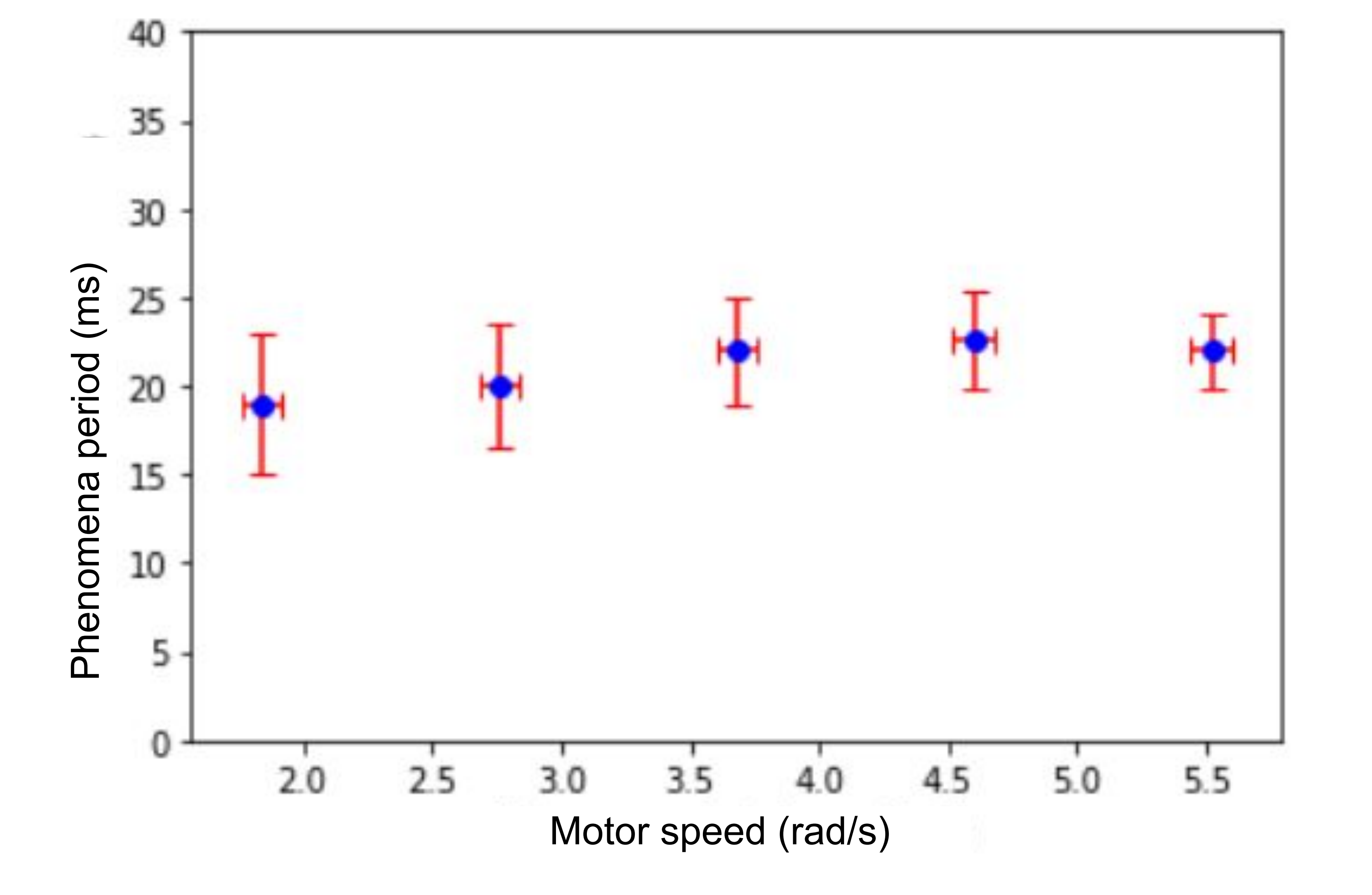
The distance between points is <u>proportional</u> to the speed

Switching to acoustic measurements with metal “chalk”



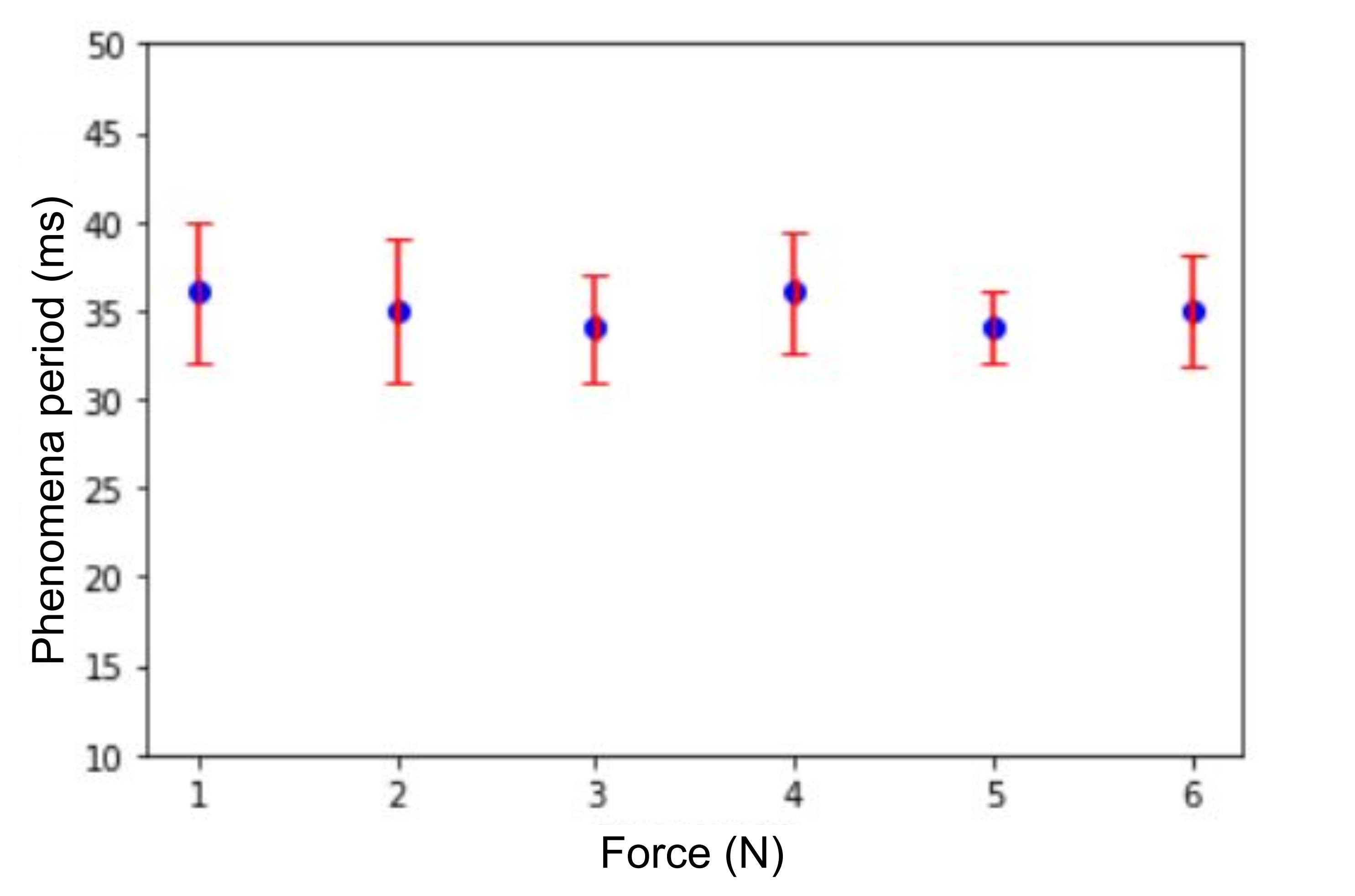


III.A – Motor speed

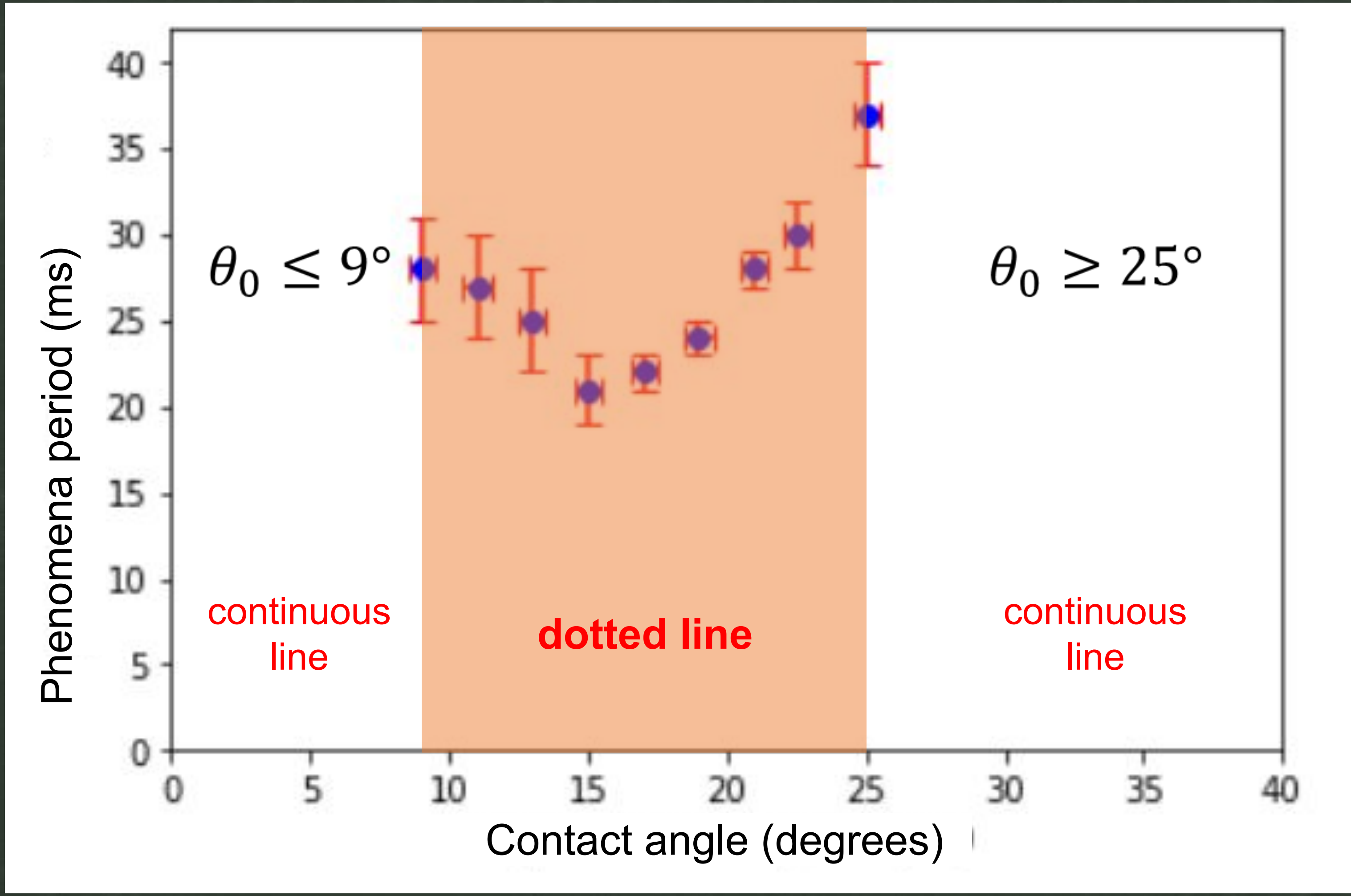


The period is <u>independent</u> from the speed

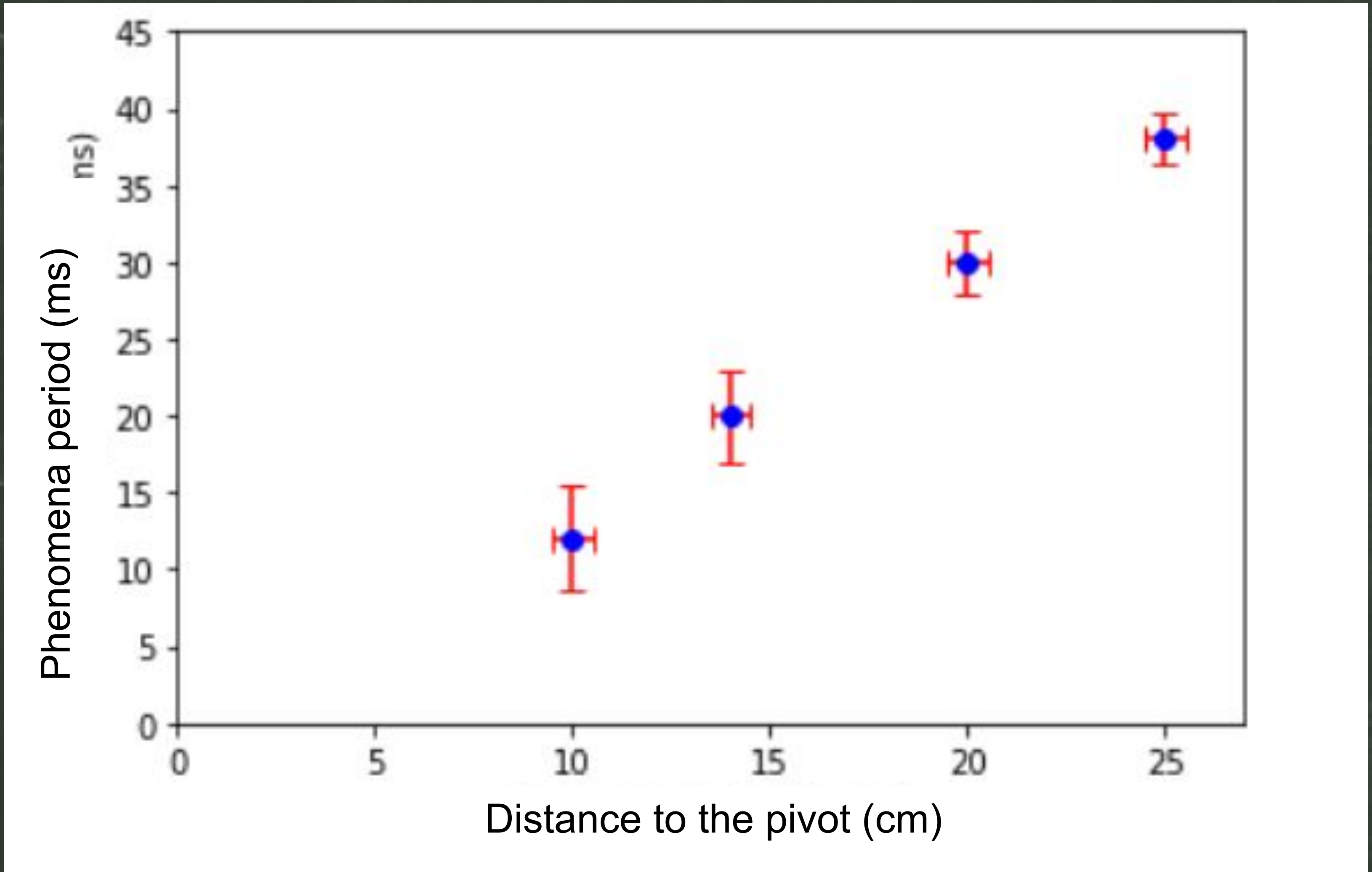
III.B – Applied force



The period is independent from the applied force

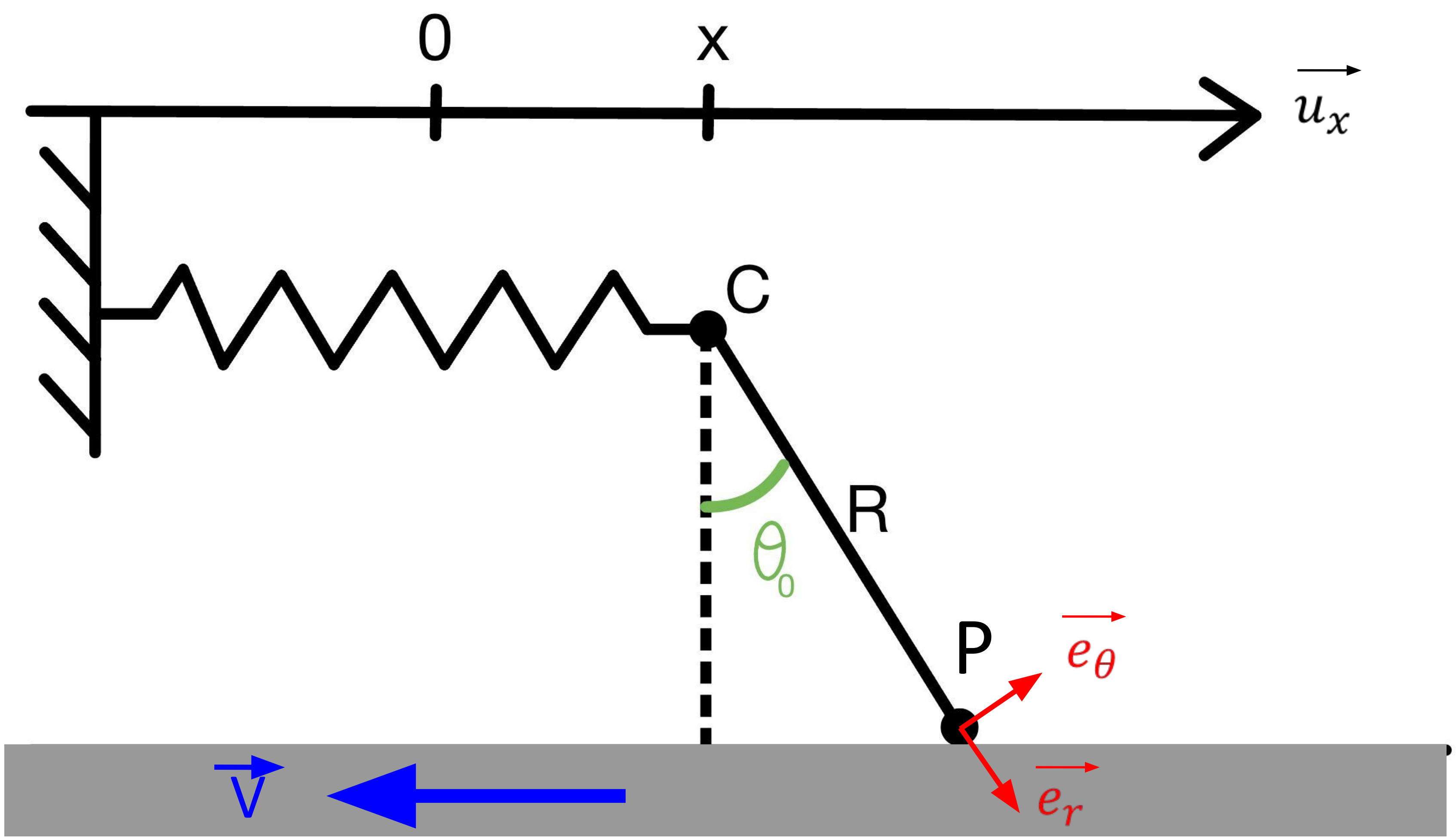


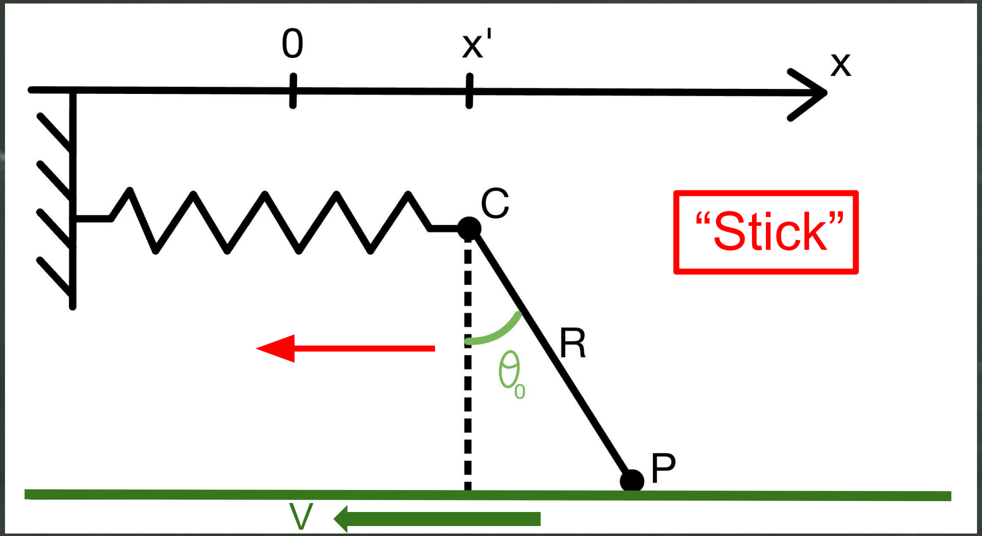
The period is <u>dependent</u> on the initial angle



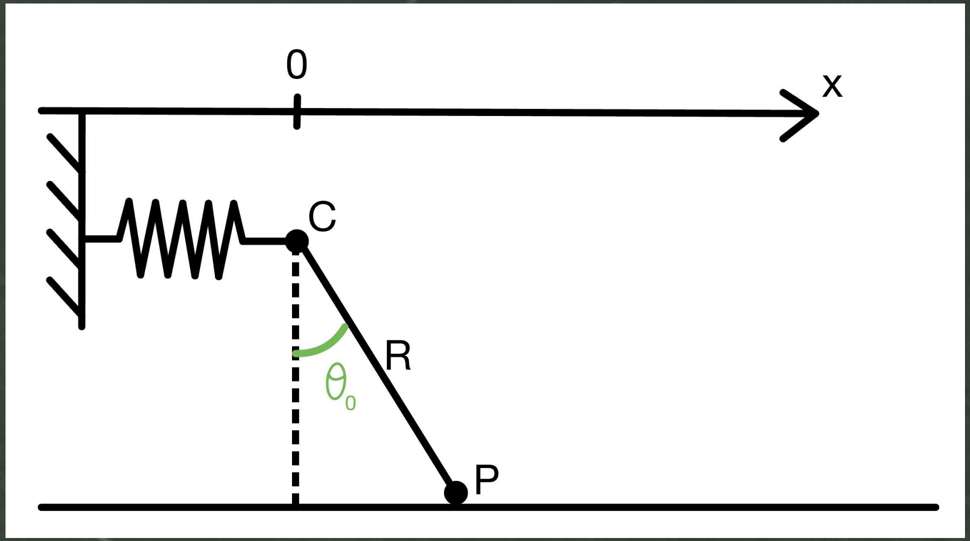
The period is <u>dependent</u> on the distance to the pivot

IV. Theoretical model

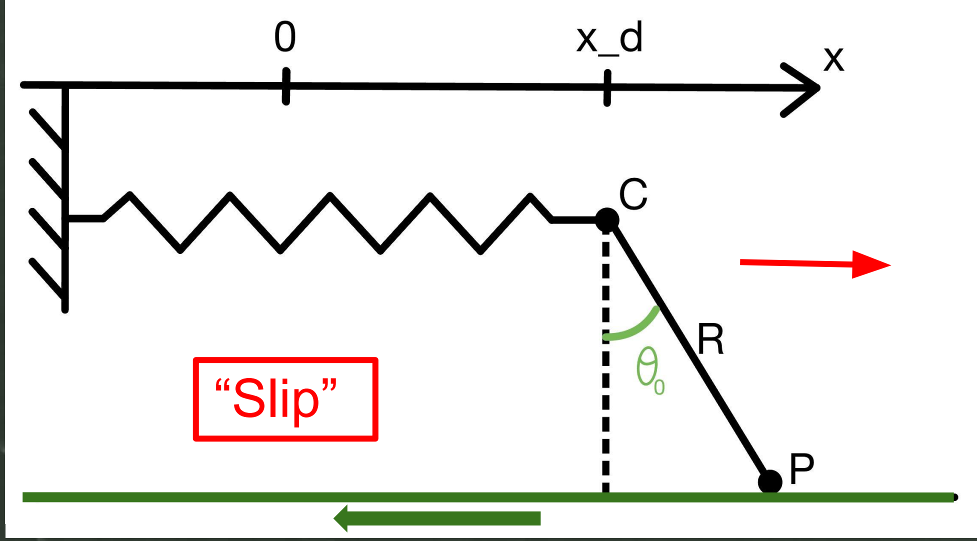




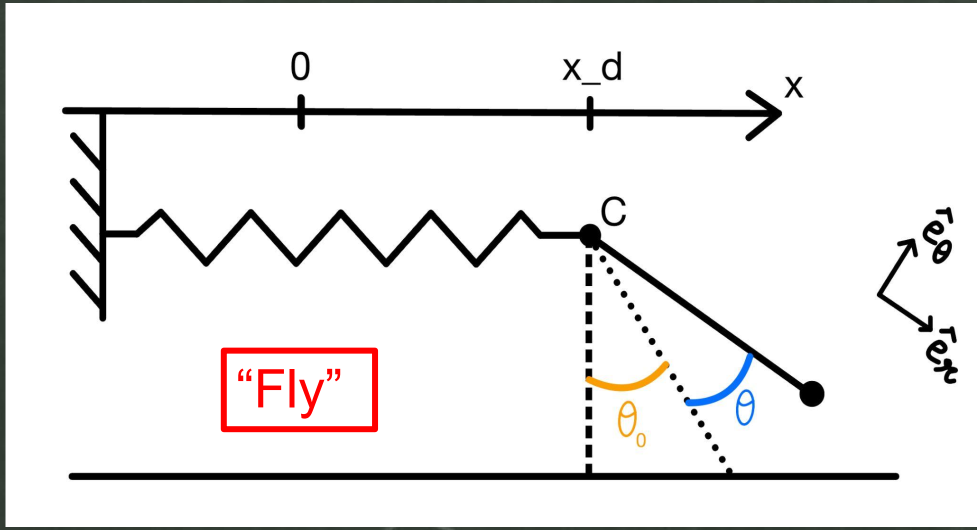
1. Spring compression



Compressed spring

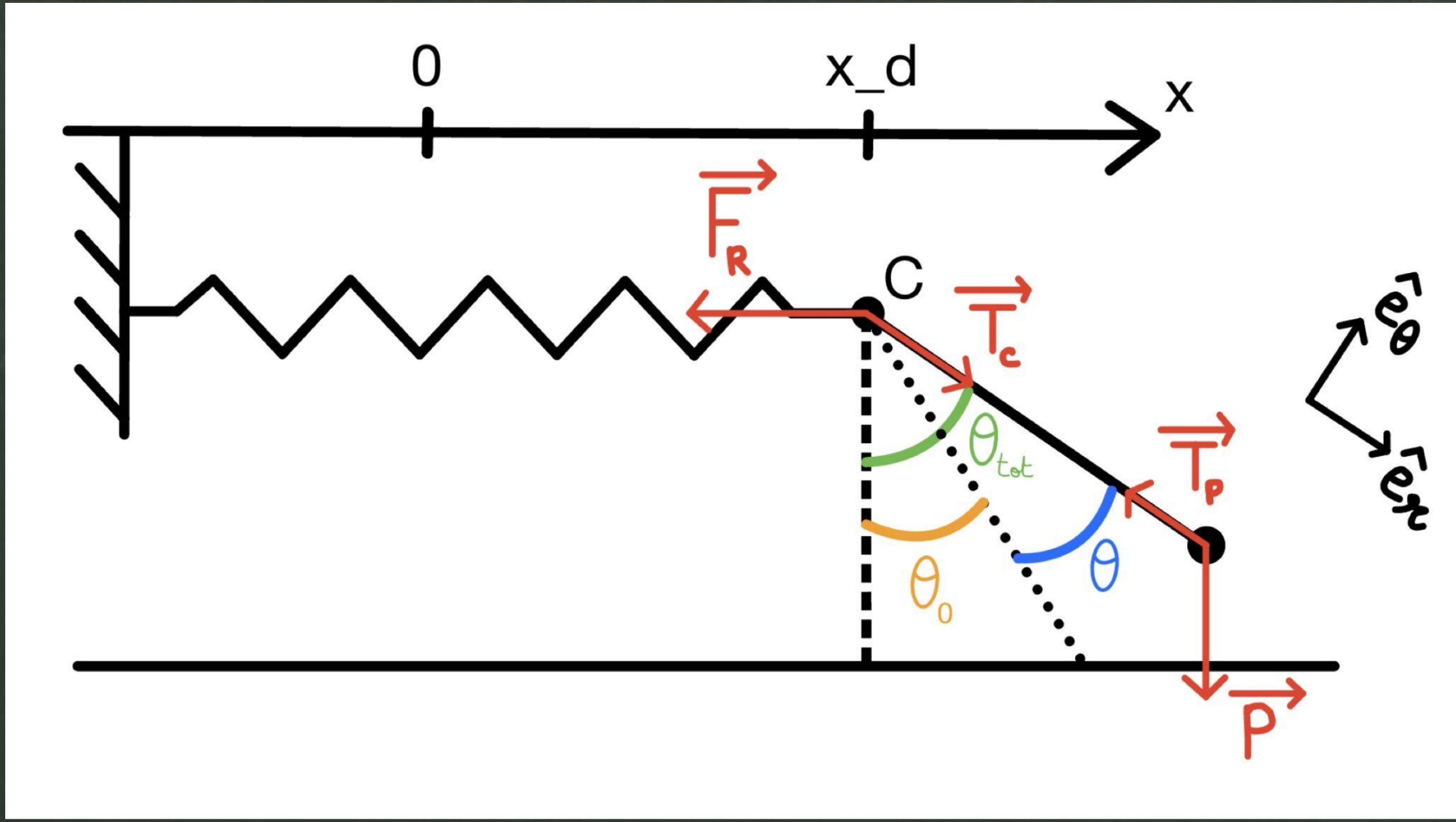


2. Spring extension



3. Flying pendulum

Summary of forces



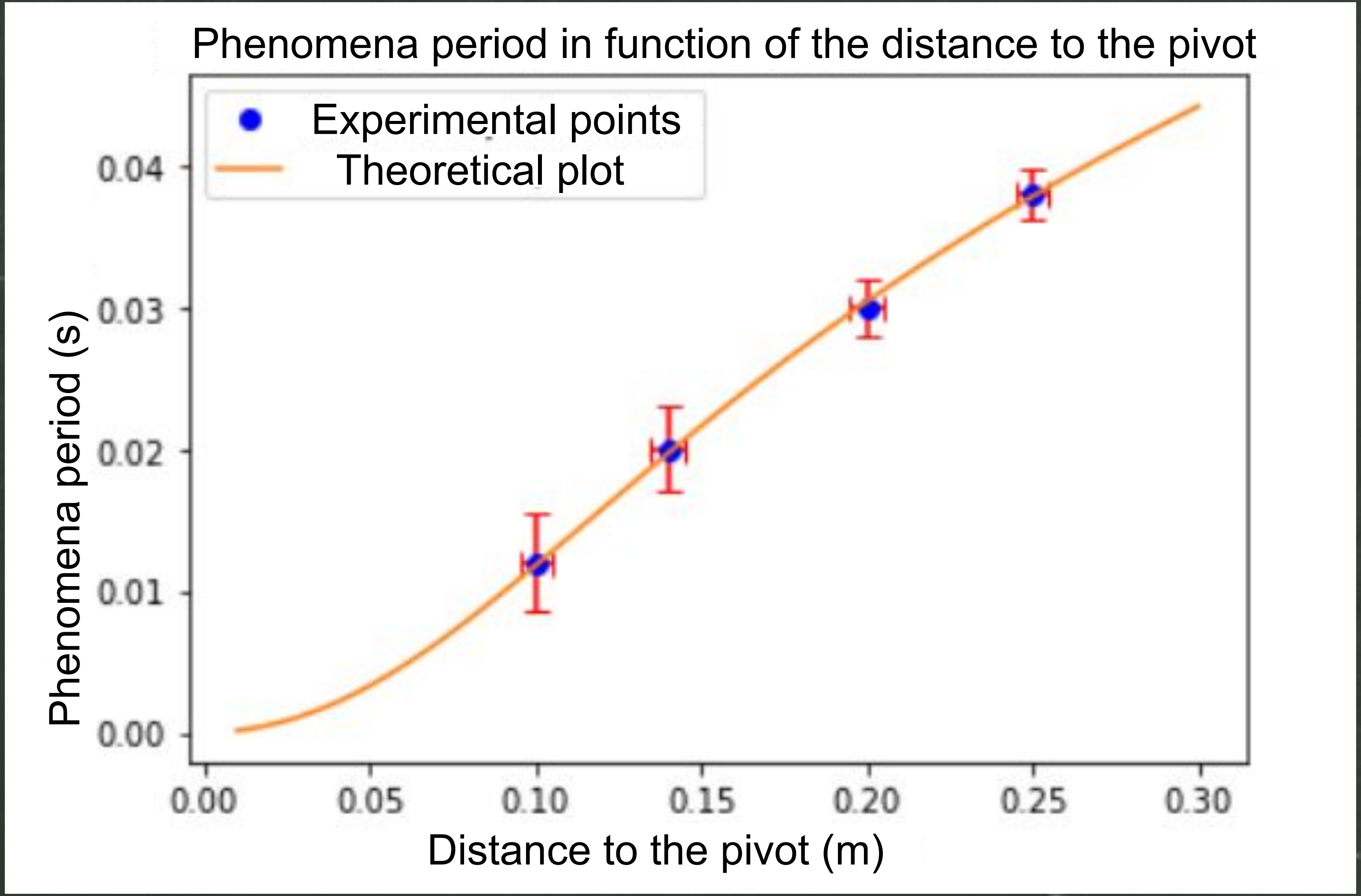
Spring force: F_R

Pendulum tension: T_P ,T_C

Weight: P

$$T = \frac{2}{\omega_0} arctan \left(\frac{(1+\frac{R}{a})\mu_s}{\theta_0} \sqrt{\frac{(1+\frac{R}{a})(mg+F)}{K}} \right)$$

$$\omega_0 = \sqrt{\frac{mg+F}{mR}}$$



$K = 43\ N.m^{-1}$ (fitted)

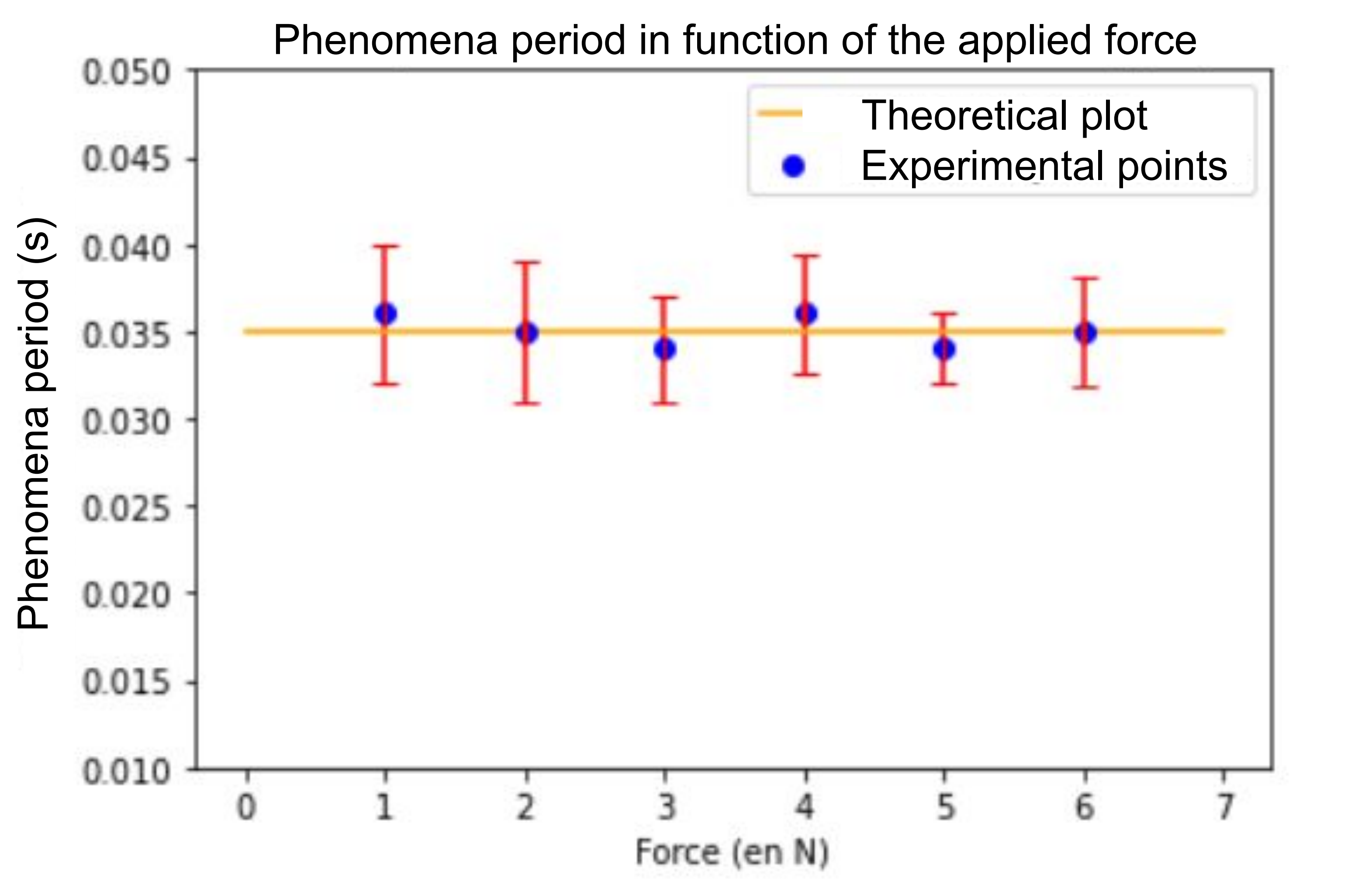
$\theta_0 = 12°$ (fixed)

$\mu_s = 0.53$ (measured)

F = 4 N (fixed)

a = 1 cm (fixed)

m = 8 kg (fitted)

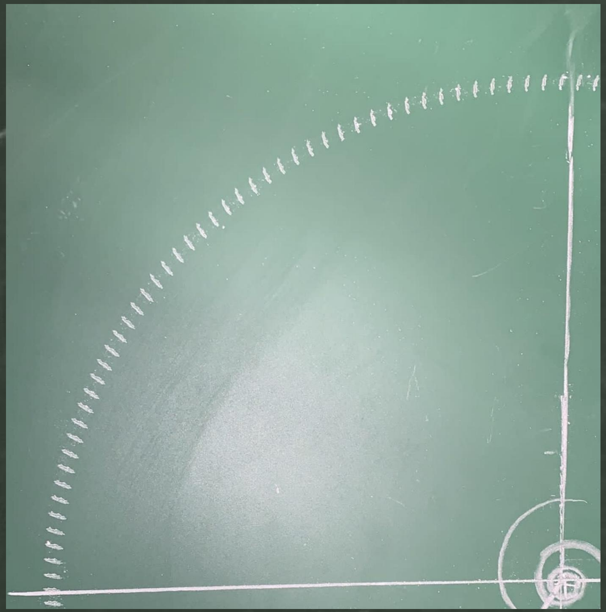


$K = 140\ N.m^{-1}$ (fitted)

$\theta_0 = 9°$ (fixed)

R = 18 cm (fixed)

m = 7 kg (fitted)



What parameters from the relative movement between the chalk and the board can be inferred from the resulting trace? Is it possible to infer anything about the dimensions of the chalk? »

→ The presence of a dotted trace can allow us to narrow the angle of contact to a range of values

→ If we compare two traces, we can make qualitative comments about the speed and the "length" of the chalk

Appendix

$$\theta(t) = (\theta_0 + tan(\theta_0))cos(\Omega_p t)$$
$$+\frac{cos(\theta_0)\omega_r L_{acc}}{Rcos\phi}\ v\ sin(\ arccos(\frac{-cos\phi}{cos^2\theta_0\mu_s}))\ sin(\Omega_p t)$$
$$-\ tan(\theta_0)$$

$$\frac{cos(\phi)}{cos^2(\theta)\mu_s} < 1$$

$$t_d = \frac{1}{\omega_r}(arccos\left(\frac{-cos\phi}{cos^2\theta_0\mu_s}\right) - \phi)$$

$$x_d = \frac{(m_1 + m_2)g\ tan(\theta_0)}{K}$$